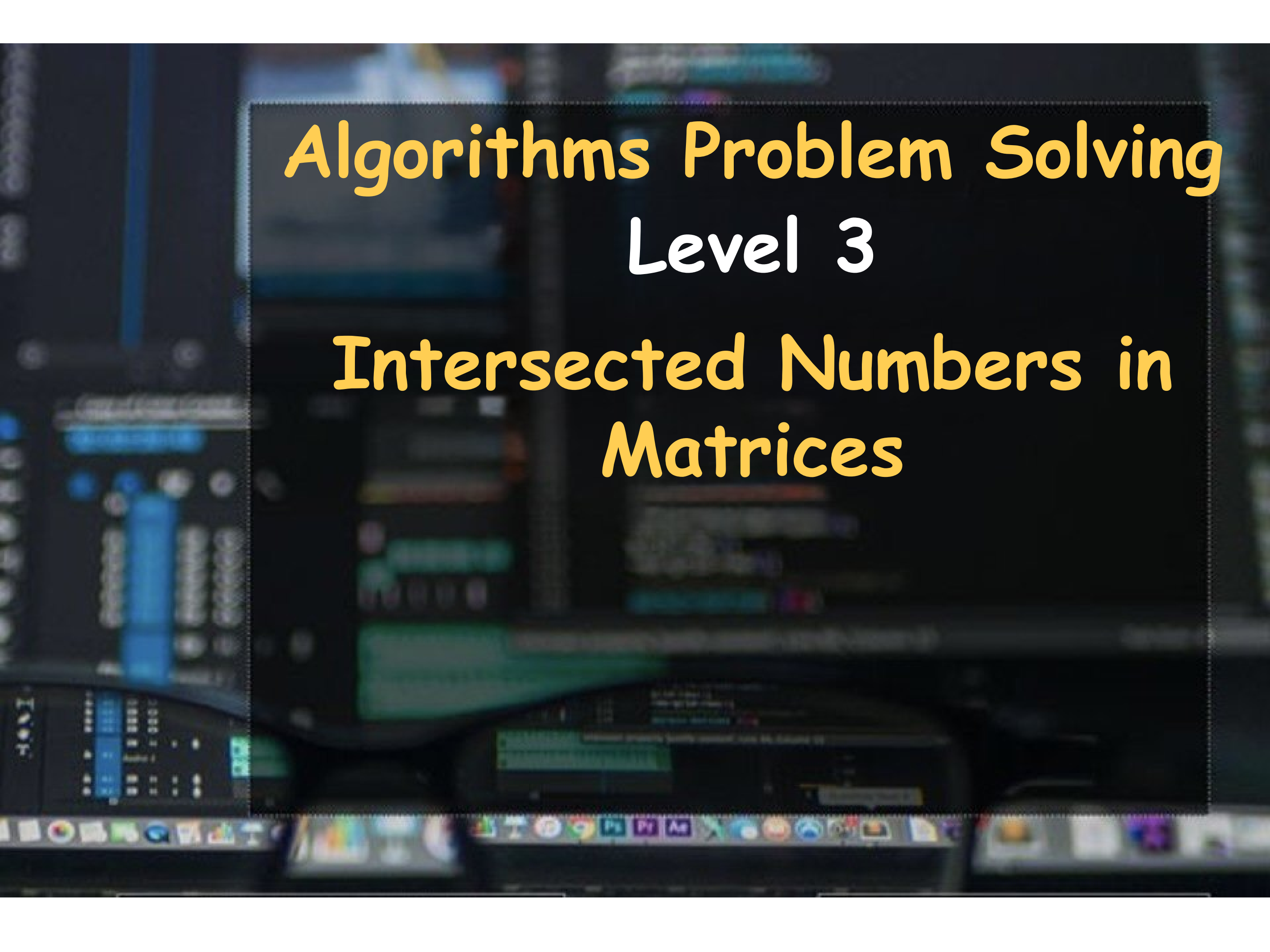
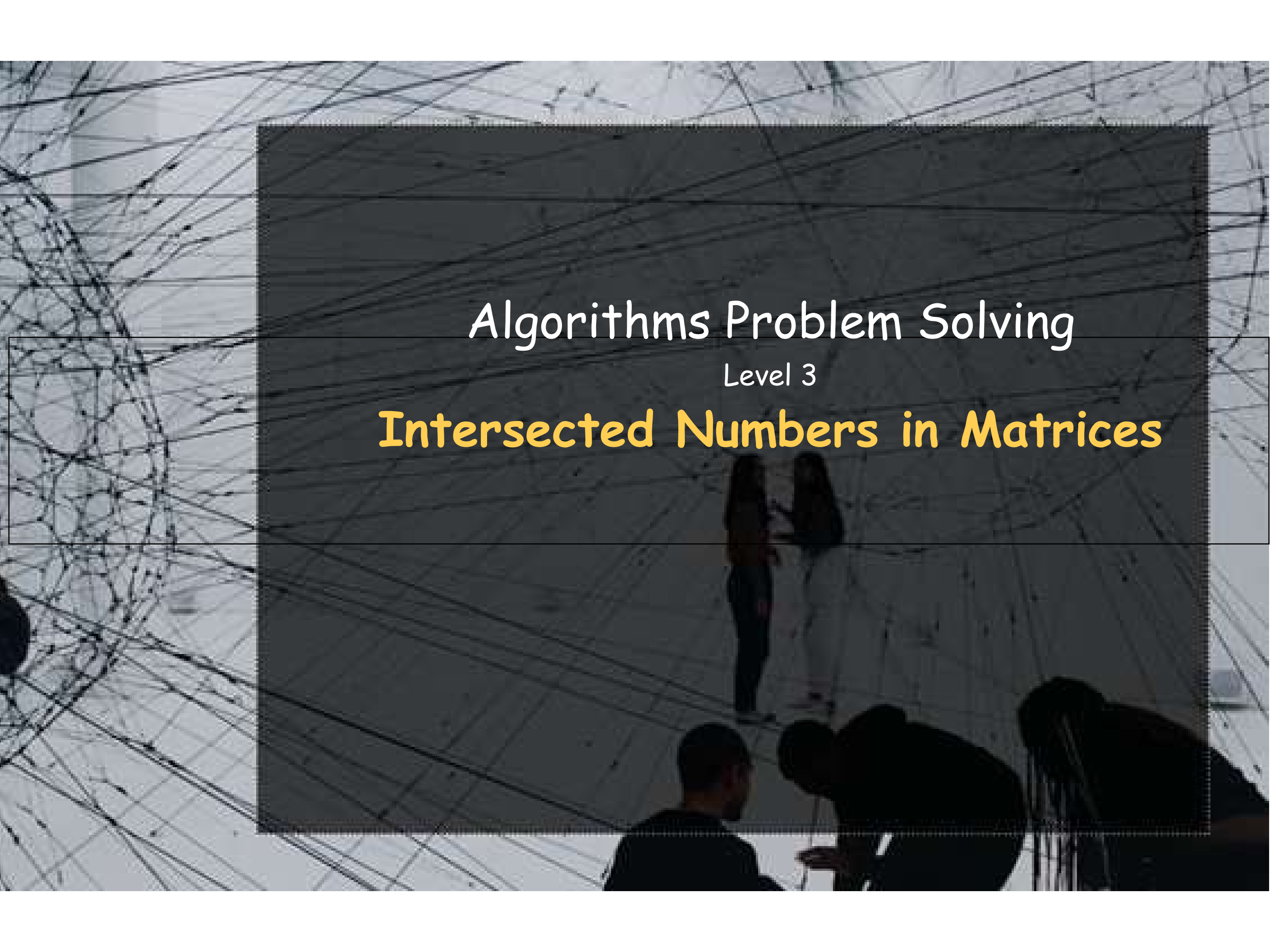




Algorithms Problem Solving Level 3

Intersected Numbers in Matrices



Algorithms Problem Solving

Level 3

Intersected Numbers in Matrices

Definition

Cross-Matrix Intersection Extraction

 Program identifies values from Matrix1 that also exist somewhere inside Matrix2.

 It demonstrates a simple set-like intersection operation using matrix traversal and membership searching.

Explanation

Matrix1 Traversal

-  Every element of Matrix1 is selected one by one.

-  The selected value becomes the search target for Matrix2.

Cross-Matrix Membership Check

-  `IsNumberInMatrix()` is used to determine whether the selected value exists in Matrix2.

-  A matching value is printed immediately.

Duplicate Behavior

-  If the same value appears multiple times in Matrix1, it can be printed multiple times.

-  Therefore, this implementation behaves more like a multiset-style traversal than a strict mathematical set intersection.

Procedures

1. Create or initialize two matrices.
2. Print both matrices.
3. Traverse every element of Matrix1.
4. Search for each value inside Matrix2.
5. Print the value when a match is found.

💡 Example

Matrix1:

77 5 12

22 20 1

1 0 9

Matrix2:

5 80 90

22 77 1

10 8 33

➡ Result

Intersected Numbers are:

77 5 22 1 1

Note

- ◆ Duplicate values from Matrix1 are preserved in the output.
- ◆ If Matrix1 contains `1` twice and Matrix2 contains `1` once, this implementation can still print `1` twice.
- ◆ The naive approach performs repeated searches and is therefore less efficient for large matrices.

⊘ Warning

- ◆ The output is not necessarily a unique mathematical set intersection.
- ◆ Time complexity can become $O((R1 \times C1) \times (R2 \times C2))$ with a linear search for every Matrix1 element.
- ◆ If unique intersection values are required, an additional duplicate-tracking mechanism is necessary.



Comparison

▨ `PrintIntersectedNumbers()`

- ✓ Checks whether each *Matrix1* value exists in *Matrix2*.
- ✓ Preserves repeated occurrences from *Matrix1*.
- ✓ Simple to implement using ``IsNumberInMatrix()``.

▤ `PrintUniqueIntersectedNumbers()`

- ✓ Prevents the same value from being printed repeatedly.
- ✓ Better represents mathematical set intersection.
- ✓ Requires additional tracking of already-printed values.

✦ Summary

- ✓ Traverses Matrix1 and searches each value inside Matrix2.
- ✓ Prints values that occur in both matrices.
- ✓ Demonstrates reuse of a boolean search helper.
- ✓ Highlights the difference between repeated-value intersection and unique set intersection.

Thank you 😊